

TABLE I
Some Benchtop Oscilloscopes

Brand	Product	Channels	Bandwidth	Sampling rate	Memory depth	Display#	Warranty	Other notable features	Price
Rigol	DS1102E/ DS1052E	Two analogue	50MHz/ 100MHz	1GSa/s	1Mpts	14.5cm (5.7-inch) TFT	Three years	- Vertical sensitivity ranges between 2mV/div and 10V/div - Standard interfaces such as USB host and device, RS232, P/F out	₹ 21,080**/ ₹ 25,564**
Tektronix	TBS1052B	Two analogue	50MHz	1GSa/s	2.5kpts	17.8cm (7-inch) TFT	Five years	34 automated measurements - Dual-window FFT - Automated, extended data-logging feature - Auto-set and signal auto-ranging - Small footprint and lightweight - Only 124.5mm (4.9-inch) deep and 2kg	₹ 32,801*
Keysight	DSO1052B	Two	50MHz	Up to 1GSa/s	Up to 16kpts	14.5cm (5.7-inch) LCD	Three years	23 automatic measurements - Waveform math including FFT - Free education student lab guides and professor slide sets - Sequential acquisition of up to 1000 trigger events, go/no-go mask testing and selectable digital filters	₹ 34,696*
Tektronix	TBS1052 B-EDU	Two analogue	50MHz	1GSa/s	2.5kpts	17.8cm (7-inch) TFT	Five years	Same as TBS1052B but comes with integrated courseware feature, which gives the ability to create lab descriptions and instructions and then upload the material directly onto a TBS1000B-EDU oscilloscope	₹ 37,911*
Tektronix	TBS1022	Two analogue	25MHz	500MSa/s	2.5kpts	14.5cm (5.7-inch) TFT	Five years	16 automated measurements - Automated, extended data-logging feature - Auto-set and signal auto-ranging - Small footprint and lightweight - Only 124.5mm (4.9-inch) deep and 2kg	₹ 39,555*
Rohde & Schwarz	HMO722/ HMO724	Two/four analogue	70MHz	Up to 2GSa/s	2Mpts	16.5cm (6.5-inch) TFT	Three years	- Mixed-signal oscilloscope with eight logic channels - Component tester, 6-digit counter, auto-measurement six parameters (maximum) - Trigger modes: Slope, video, pulsewidth, logic, delayed and event	N/A
Scienteck Technologies	Scienteck 401	One	50MHz	500MSa/s	32kpts	17.8cm (7-inch) TFT	One year	- Real-time sampling rate 500MSa/s, equivalent sampling rate 50GSa/s - Unique digital filter function and waveform recorder function 32 parameters auto-measure function	N/A
Scienteck Technologies	Scienteck 402	One	70MHz	1GSa/s	2Mpts	17.8cm (7-inch) TFT	One year	With a higher bandwidth, memory depth and sampling rate, other notable features are same as above	N/A

*Price listed as on e-commerce websites like element14.com and others

Colour displays

** Price converted from US\$ to ₹ (1 US\$ = ₹ 64). Landing price may vary slightly when bought from Indian distributors

design houses, providing affordable performance on a tight budget. Although, these limit mobility and consume considerable amount of bench space, most oscilloscopes available under this category come with reliable features required for basic T&M needs such as USB connectivity for easily connecting to a laptop or per-

sonal computer and data storage purposes, signal auto ranging, numerous automated measurements, automated data logging, frequency counter, auto set with selective bandwidth models like 25MHz, 40MHz, 50MHz, 60MHz, 70MHz, 100MHz, 150MHz and 200MHz.

Considering a price point of

₹ 50,000 and below, majority of the general-purpose digital scopes offer bandwidths up to 70MHz with dual analogue channels. The sample rate for such oscilloscopes is usually between 1GSa/s and 2GSa/s. Higher bandwidth and four analogue channels for educational and industrial purposes exceed the ₹ 50,000 mark.